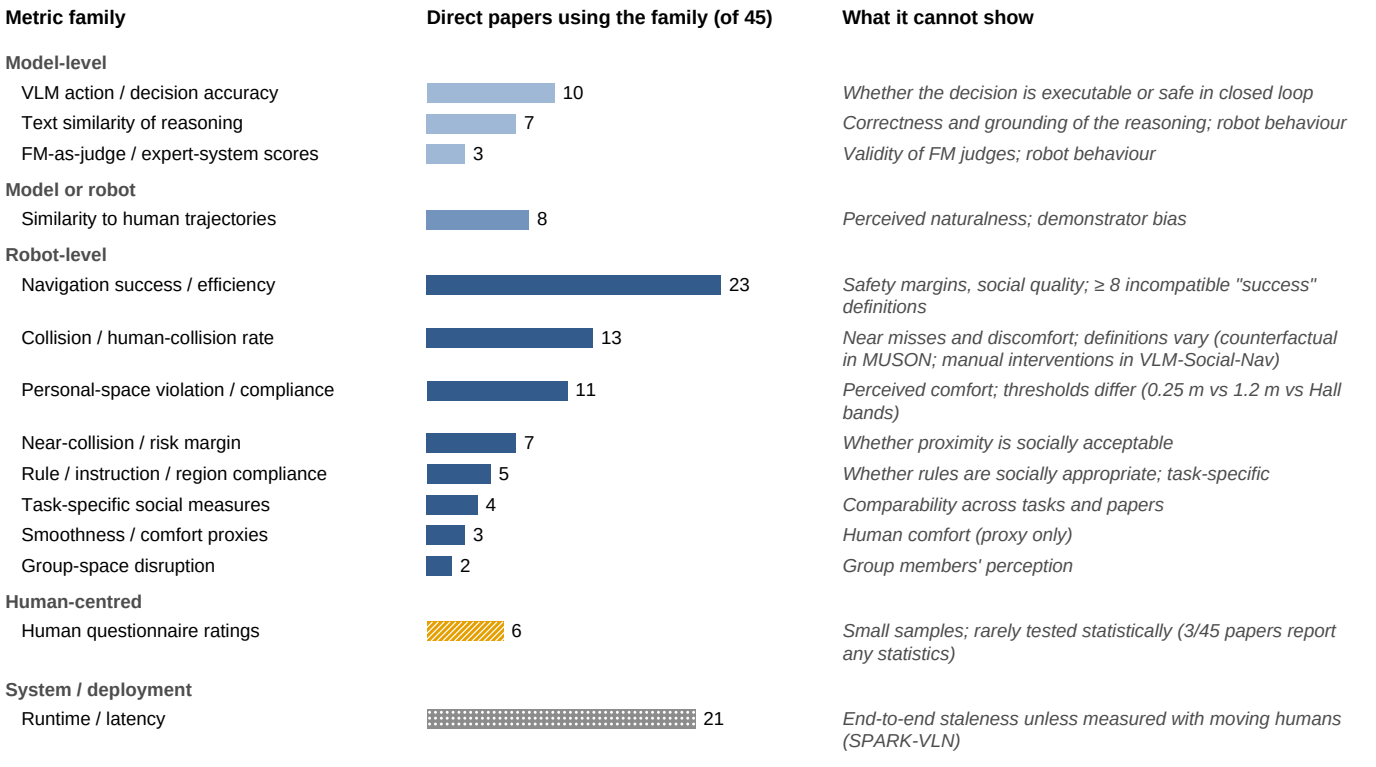
Metric family
Direct papers using the family (of 45)
What it cannot show
Model-level
VLM action / decision accuracy
10
Whether the decision is executable or safe in closed loop
Text similarity of reasoning
7
Correctness and grounding of the reasoning; robot behaviour
FM-as-judge / expert-system scores
3
Validity of FM judges; robot behaviour
Model or robot
Similarity to human trajectories
8
Perceived naturalness; demonstrator bias
Robot-level
Navigation success / efficiency
23
Safety margins, social quality; ≥ 8 incompatible "success" definitions
Collision / human-collision rate
13
Near misses and discomfort; definitions vary (counterfactual in MUSON; manual interventions in VLM-Social-Nav)
Personal-space violation / compliance
11
Perceived comfort; thresholds differ (0.25 m vs 1.2 m vs Hall bands)
Near-collision / risk margin
7
Whether proximity is socially acceptable
Rule / instruction / region compliance
5
Whether rules are socially appropriate; task-specific
Task-specific social measures
4
Comparability across tasks and papers
Smoothness / comfort proxies
3
Human comfort (proxy only)
Group-space disruption
2
Group members' perception
Human-centred
Human questionnaire ratings
6
Small samples; rarely tested statistically (3/45 papers report any statistics)
System / deployment
Runtime / latency
21
End-to-end staleness unless measured with moving humans (SPARK-VLN)